Problem Set 7:

1) Given: T is a linear operator

a). To prove: T has a 1-dimensional invariant iff T has an eigenvalue.

b). range(T) = invariant subspace of T.

c). Nullspace(T) is invariant subspace of T.

T has eigenvalue λ, what about Nullspace $(T - \lambda I)$.

Proof:

Let W be a 1-dimensional invariant subspace of T.

$$W = \text{span}\{v\}.$$

$$T(v) \in W.$$

$\therefore T(v)$ must be a scalar multiple of ~~W~~ v.

Page No.
Date

$\therefore T(v) = \lambda v$.

$\therefore$ v is an eigenvector
λ is an eigenvalue.

(ii) <u>backwards proof</u>:

$T(v) = \lambda v$

and $W = \text{span}(v)$.

$\therefore$ for $w \in W$

$w = av$

$T(w) = T(av) = aT(v) = (\lambda v)a$
$= \lambda(av)$

$\therefore T(w) = \lambda w$.

$\therefore T(w) \in W$.

$\therefore W$ is an invariant

b). $T(\text{range}(T)) \subseteq \text{range}(T)$.

$y \in \text{range}(T)$.

$y = T(x)$.

$T(y) = T(T(x))$

$T(y) = T^2(x)$.

$T(y) \in \text{range}(T)$

$T(\text{range}(T)) \subseteq \text{range}(T)$.

$\therefore$ range (T) is invariant

c). $T(\text{null}(T)) \subseteq \text{null}(T)$.

$v \in \text{null}(T)$.

$\therefore T(v) = 0$.

$T(T(v)) = T(0)$.

$T(T(v)) = 0$

$T(v) \in \text{null}(T)$

$\therefore T(\text{null}(T)) \subseteq \text{null}(T)$.

$\therefore$ nullspace (T) is invariant

d). $T(v) = \lambda v$

$T(v) - \lambda v = 0$

~~$T(v$~~ $v(T - \lambda I) = 0$.

Page No.
Date

$\therefore v \in null(T - \lambda I)$.
$null(T-\lambda I) \neq \{0\}$.

$\therefore null(T-\lambda I)$ is eigenvector corresponding to eigenvalue λ.

it's eigenspace of T corresponding to λ.

2). Given: $S, T \in L(V)$ be linear operators.

To prove: nullspace(S) and range(S) are invariants of T.

Proof:

As S and T commute,

$ST = TS$.

i). for W invariant under T if $v \in W$,

$T(v) \in W$.

Page No.
Date

$v \in null(S)$.

$S(v) = 0$.

$S(T(v)) = T(S(v))$

But, $S(v) = 0$.

$T(S(v)) = T(0) = 0$.

$\therefore S(T(v)) = 0$

$T(v) \in null(S)$.

$\therefore T(null(S)) \subseteq null(S)$.

$\therefore$ null(S) is invariant under T.

ii) $w \in range(w)$
$w = S(v)$.

$T(w) = T(S(v)) = S(T(v))$

$\therefore T(w) \in range(S)$.

$\therefore T(range(S)) \subseteq range(S)$.

$\therefore$ range(S) is invariant under T.

3). $T: P(\mathbb{R}) \to P(\mathbb{R})$ by $T(p) = p'$

To find: eigenvalues & corresponding eigenvectors of T.

Sol^n: if there's a non-zero polynomial $p(x)$

$$T(p) = \lambda p$$

if λ is eigenvalue

$$\therefore T(p) = p' = \lambda p.$$

$$\therefore \frac{dx}{dy} = \lambda x$$

$$\int \frac{dx}{x} = \int \lambda \, dy$$

Page No.

Date

$$p'(x) = \lambda p(x)$$

$$\int \frac{p'(x)\,dx}{p(x)} = \int \lambda\, dx$$

$$\therefore p(x) = Ce^{\lambda x}$$

$$\lambda = 0.$$

$$\therefore \quad p'(x) = 0$$

$$\therefore p(x) = c$$

$\therefore$ All non zero constant polynomials are eigenvectors

$\therefore$ eigenspace is

span $\{1\}$,

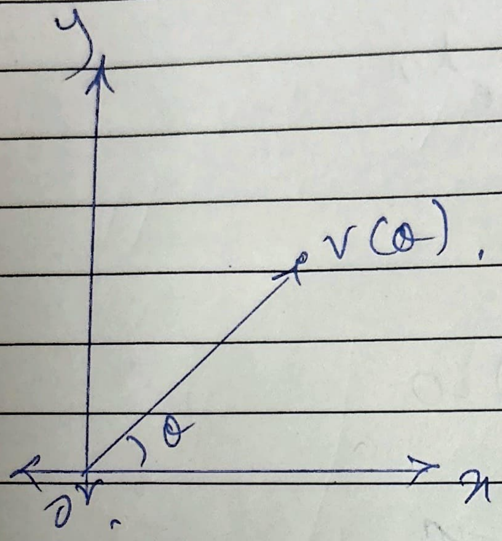
$$\therefore \{p(x) = c \mid c \in \mathbb{R}\} = \text{span}\{1\}.$$

4) Given: $V = R^2$

$v(\theta) \in V$

$v(\theta) = r(\cos\theta, \sin\theta)$.

Soln:

a) 

b). $T(v(\theta)) = r(\cos(\theta+\phi), \sin(\theta+\phi))$

as we rotate it counterclockwise at angle ϕ.

c). $\phi = \pi/2$

$\cos(\theta + \pi/2) = -\sin\theta$

$\sin(\theta + \pi/2) = \cos\theta$

Date

$\therefore T(v(\theta)) = r(-\sin\theta, \cos\theta)$.

$\therefore v(\theta) = (r\cos\theta, r\sin\theta) = (a, b)$

$\therefore a = r\cos\theta$
$b = r\sin\theta$.

$\therefore T(v) = (-b, a)$

d) $T(v) = \lambda v$.

$\therefore$ Transformation SCALES a vector, NOT its direction.

$\therefore$ its on the SAME LINE as original vector.

But, $\phi = \pi/2$

$T(v) = (-b, a)$

$\therefore Tv \neq \lambda v$.

for $\lambda \in R$ (if $v \neq 0$).

5) Proof :

for

(a) => (b).

T is with respect to $v_1 \ldots v_n$ is upper triangular.

$$\therefore [T]_\beta = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & 0 & \ddots & \vdots \\ 0 & \cdots & \cdots & a_{nn} \end{bmatrix}$$

$$\therefore T(v_j) = a_{1j}v_1 + \cdots + a_{jj}v_j$$

$$\therefore T(v_j) \in \operatorname{span}(v_1, \ldots v_j).$$

Page No.

Date

for (b) $\Rightarrow$ a.

$T(v_j) = \text{span}(v_1, \ldots v_j)$

$\therefore T(v_j) = a_{1j} v_1 + \ldots + a_{jj} v_j$.

$\therefore a_{(j+1)j} = a_{(j+2)j} = \ldots = 0$.

$\therefore$ it's an upper Δ matrix.

for (b) $\Rightarrow$ (c).

$T(v_j) \in \text{span}(v_1, \ldots v_j)$

$W_j = \text{span}(v_1 \ldots v_j)$.

$v = a_1 v_1 + \ldots + a_j v_j$

$T(v) = a_1 T(v_1) + \ldots + a_j T(v_j)$

$\therefore T(v) \in W_j$

$T(W_j) \subseteq W_j$

$\therefore W_j$ is invariant under T.

Page No.

Date

$(c) \Rightarrow (b)$

$span(v_1, \ldots, v_j)$ is invariant under T

$\therefore v_j \in span(v_1, \ldots v_j)$

$\therefore T(v_j) \in span(v_1, \ldots v_j)$.

$\therefore b$ is true.

$\therefore a \Leftrightarrow b \Leftrightarrow c$.

6) $T: F^2 \to F^2$.

$T(x, y) = \cancel{0(x)}(0, x)$

Let $U = \{(0, y), y \in F\}$

a) $u \in U$,

$u = (0, y)$, $y \in F$

$T(0, y) = (0, 0)$.

But $0(y) = 0(0, y) = (0, 0)$.

$\therefore T(u) = 0(u)$.

$\therefore$ ~~There~~ All non-zero vectors in U are eigenvectors and eigenvalue is 0.

b) $T(x, y) = \lambda(x, y) = (0, x)$

$\therefore (\lambda x, \lambda y) = (0, x)$

$\therefore 0 = \lambda x$

$\lambda y = x$

$\therefore 0 = \lambda y$

$\boxed{\cancel{y=0}}$ $y \neq 0$

$\therefore \boxed{\lambda = 0}$.

~~But we~~

$\therefore$ only eigenvalue is 0,

c). Let $v = (x, y)$ be an eigenvector,

$T(v) = 0v = (0,0)$

$T(x,y) = (0,x) = (0,0)$

$x = 0$

$v = (0,y)$

$U = \{(0,y) : y \in F\}$

$\therefore$ every eigenvector lies in U.

d). $U = \{(0,y)\}$ is 1-dimensional

If $F^2 = U \oplus W$,

$\dim(W) = 1$

Page No.
Date

$\therefore$ W must be spanned by a non zero vector ~~a~~ a

But W is invariant,

$\therefore$ a must be an eigenvector

$\therefore$ a is eigenvector of T

$\therefore a \in V$

$\therefore W \subseteq V$

$\therefore V \cap W \neq \{0\}$

$\therefore$ Proof by contradiction,

there's no subspace W such that

$F^2 = V \oplus W$ & W is invariant under T.

Page No.

Date

7).

a). Let $v, Tv, T^2v \dots T^nv$ be in V.

$\dim(V)$ is linearly dependent.

$\therefore \exists\, a_0, a_1, \dots a_n$ aren't all zeroes.

So, $a_0v + a_1Tv + \dots + a_nT^nv = 0$

$$\therefore \sum_{i=0}^{n} a_i T^i v = 0.$$

Let $a_0 = 0$.

$\therefore a_1Tv + \dots + a_nT^nv = 0$.

$\therefore T(a_1v + a_2Tv + \dots + a_nT^{n-1}v) =$

$w = a_1v + a_2Tv + \dots + a_nT^{n-1}v$.

$\therefore$ ~~$T(w) = 0$~~.

$T(w) = 0$

$w \in \text{null}(T)$.

~~as v, Tv, T^2v - - was chosen as a m.~~

as part (a) was chosen as a minimal relation among $v, Tv, T^2v, \ldots$

so coefficient of v cannot be 0.

$\therefore a_0 \neq 0$,

c). $c(T-\lambda_1 I) \ldots \cancel{T-\lambda_m I}$ $(T-\lambda_m I)v = 0$.

$c(T-\lambda_1 I) \ldots (T-\lambda_m I)v = 0$.

$c \neq 0$.

$w = (T-\lambda_2 I) \ldots (T-\lambda_m I)v$.

$\therefore (T-\lambda_1 I)w = 0$

$w \neq 0$.

$\therefore w$ is in nullspace of $T-\lambda_1 I$.

$\therefore$ its NOT injective

$\therefore$ for some j,

$$(T - \lambda_j I)u = 0$$

$\therefore$ $T - \lambda_j I$ isn't injective.